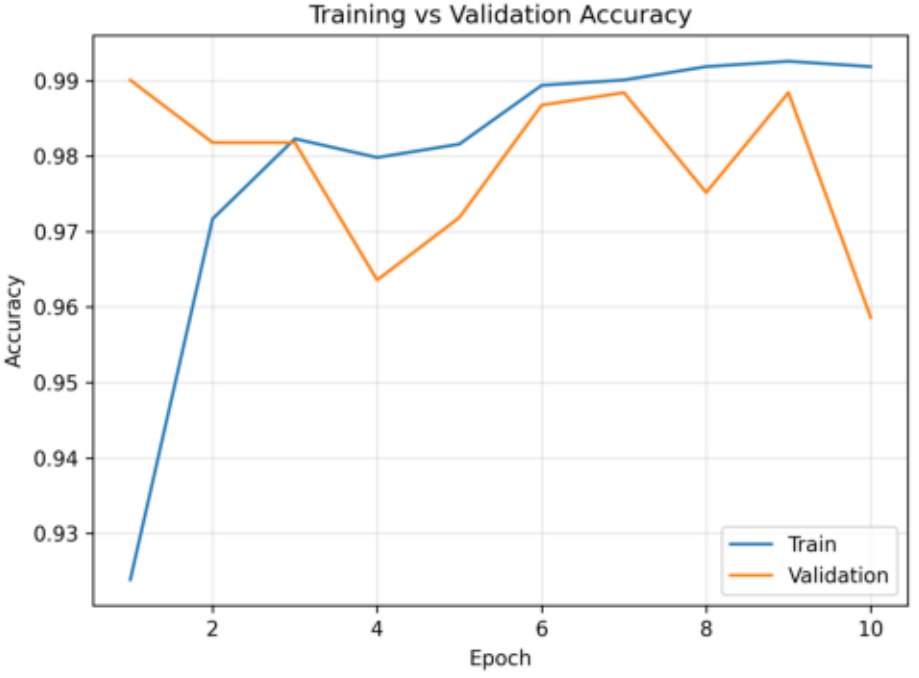
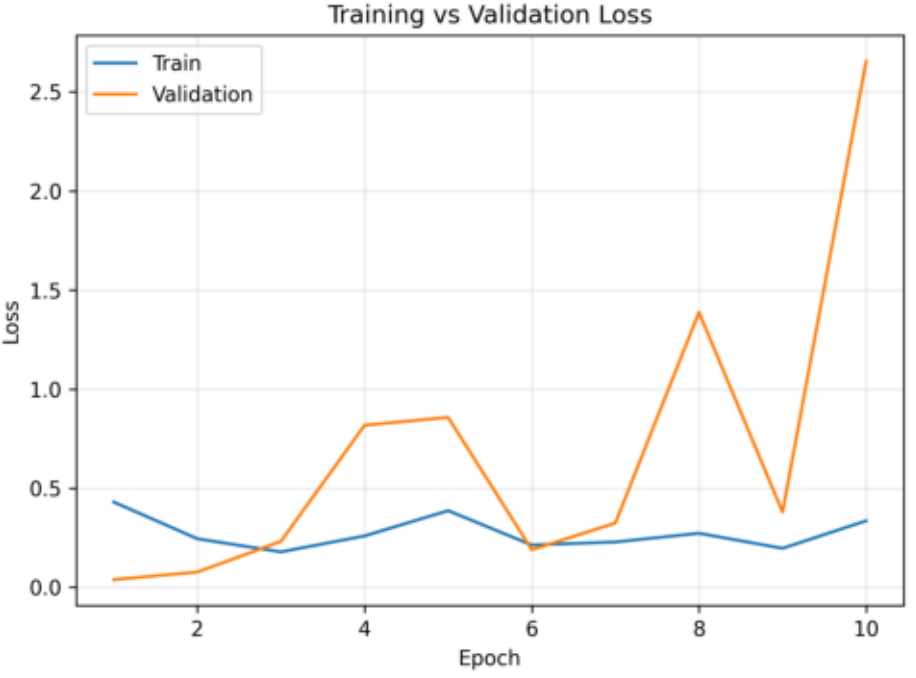
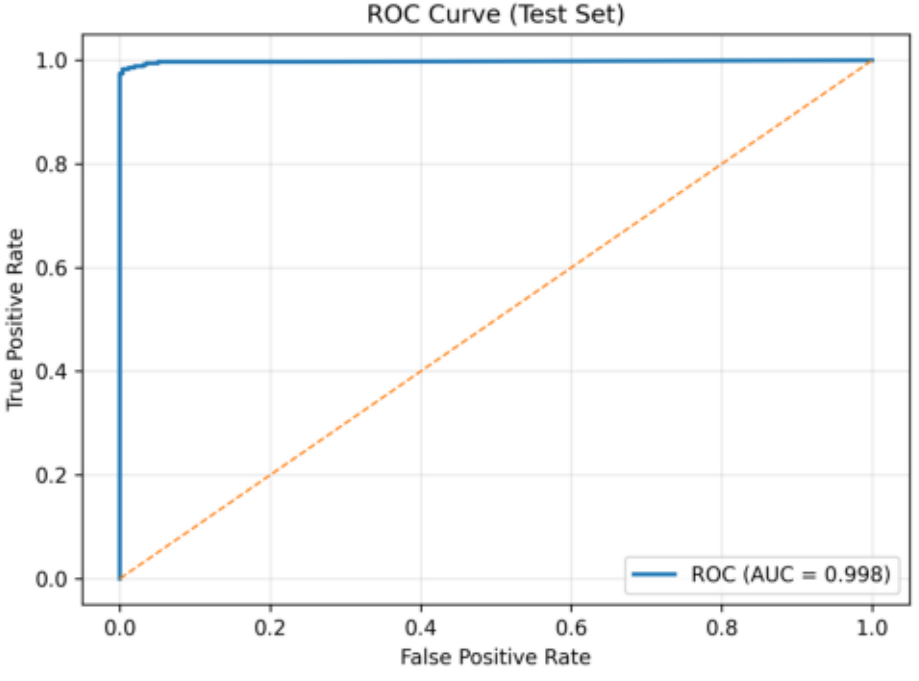
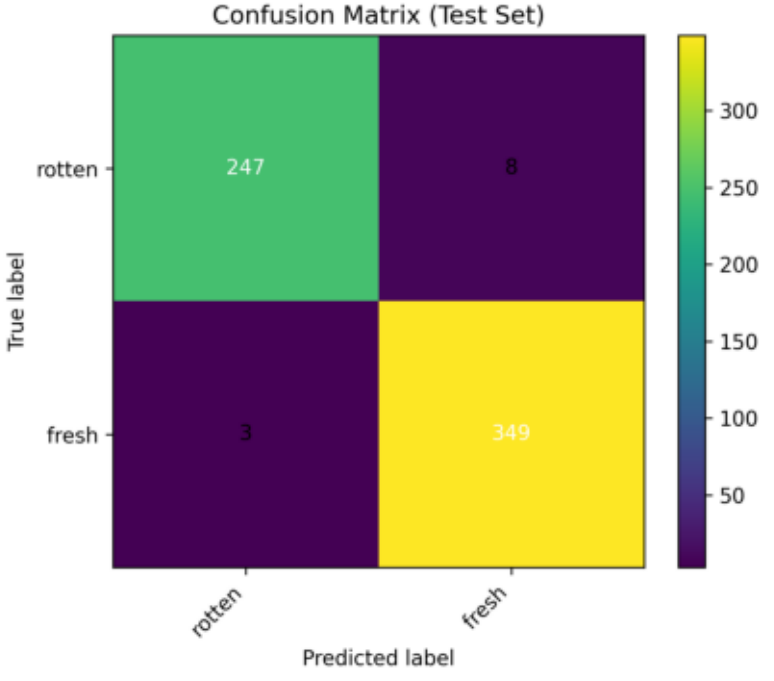
VGG16 Apple Freshness Classification — Results Summary

```
Summary Metrics (Test Set):
VGG16 (frozen conv backbone) | Test | Acc=0.9819  Prec=0.9776  Rec
```


Training vs Validation Accuracy
Accuracy
Epoch
0.99
0.98
0.97
0.96
0.95
0.94
0.93
2
4
6
8
10
Train
Validation


Training vs Validation Loss
Train
Validation
Loss
Epoch
0.0
0.5
1.0
1.5
2.0
2.5
2
4
6
8
10


ROC Curve (Test Set)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC (AUC = 0.998)


Confusion Matrix (Test Set)
rotten
fresh
True label
247
8
3
349
rotten
fresh
Predicted label
300
250
200
150
100
50

Comparison Table

Model	Split	Accuracy	Precision	Recall	F1	AUC
VGG16 (frozen conv backbone)	Test	0.9819	0.9776	0.9915	0.9845	0.9988